

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

BEHAVIOR AND MANAGEMENT OF SHEEP & GOAT

Precautions to be taken before serving:

For Ram:

- **Selection of breeding males:** Selection should be conducted to choose the males that used for the breeding season, according to
 - 1.their fitness strength.
 - 2.Seminal quality
 - 3.freedom from venereal diseases.

In the tuppig season, **3 rams are left with 100 ewes** for as long as three months.

- **Shearing:** before the breeding season to increase their activities.

Precautions to be taken before serving:

For Ram:

- **Trimming of ram claws** is important to facilitate serving.
- **Flushing:** Over fat rams should kept on the same feeding programs while, **thin one** should be flushed by increase the daily provision of **500 gm of concentrate mixture / head / day.**
- **Exercise:** in the form daily walking for **4 - 6** kilometer.

Precautions to be taken before serving:

For Ram:

- **Marking and identification:** To differentiate between rams to know which are serviced ewes, paint each by special coloring paint every week. In such cases, serviced ewes are marked with a spot of paint or marking fluid and in this case record the **date of service**. Usually services occur at night or early morning.
- **Parasitic control:** Rams should be treated with suitable **anthelmintic** for internal parasites and external parasites.

Precautions to be taken before serving:

For EWES:

- **Culling:** Only **high-producing ewes** should be kept. When the flock owner grows his own replacement should save each year from **1/5 to 1/6 ewe lambs** from heavy-shearing good milking dams.
- **Shearing:** before breeding season to increases the activity and the vitality of ewes and prevents summer infertility.
- **Dipping** or spraying of ewes with insecticide to treat **external parasites**.

Precautions to be taken before serving:

For Ewe:

- **Care of foot (Trimming of Claws):** preferable on grass fields.
- **Examination and testing:** against venereal diseases.
- **Flushing:**
 1. The good condition ewe usually, conceives at the first service and frequently produces twin lambs.
 2. The thin ewe or excessively fattening is difficult to get in lamb and seldom produces twins.

Precautions to be taken before serving:

For Ewe:

- Procedure: Two weeks previous to and during mating season give the flock the best pasture on the farm. A change to **fresh pasture** increases the quality of the ration. When it is not possible to provide improved pasture, the ewes should be fed from **1/2 to 3/4 pound of grain per head daily**.

Precautions to be taken before serving:

For Ewe:

- **Synchronization of estrus:** this can be done by **synthetic progesterone** where a pessary or spongy which is impregnated with a synthetic progesterone compound into vagina of the ewe which slowly release progesterone. Sponges are left in the ewes for approximately the same time, as the **CL (Corpus Lutiem)** is normally present on the ovary **(12-14 days)**. When the sponges are withdrawn from ewes all of them come in estrus.

Pregnancy diagnosis:

- 1.By Ultrasound or Radiography:** used to see the fetus skeleton and heartbeats from the **43rd day of gestation**.
- 2.By abdominal palpation:** During **late stage of pregnancy** the fetus could be palpated in front the pelvic cavity.
- 3.Vaginal examination :**by **speculum** to see the plug formed by cervical gland secretion. This plug closed the cervical canal during pregnancy period and prevents entrance of infection to the fetus and then solved before parturition.
- 4.Service examination:** by putting serviced ewes with **teaser rams for 35 days**, the pregnant ones do not show signs of estrus.

Pregnancy diagnosis:

5. By signs of pregnancy such as:

1. Cessation of estrus.
2. Improvement in healthy condition and increase of abdominal size.
3. Development of udder.
4. Falling in flank region.
5. Slackening of pelvic ligaments.
6. Mucous discharges from vulva just before

Management during pregnancy:

- 1. Feeding:** Pregnant ewes should be feed an **easily digestible ration** especially during the last half of pregnancy by grazing in the fields and **avoid excess of starchy or watery food.**
- 2. Exercising:** Pregnant ewes should be exercised daily for **not more than 2 km** and avoid sudden disturbances & crowding.

Management during pregnancy:

3. Housing: Pregnant ewes may lamb in the field or in **lambing pen**. Lambing pen is small box about **1.5 square meters**, suitable to hold **one ewe and her lambs**. **Wheat straw or rice straw** is preferred for parturition and care is taken to keep the pens **clean & disinfected**. The individual pens are provided with **infrared heaters**, which can be used for weakly lambs. Ewe should stay in the pen for **48 hours** before being moved outside to a field.

Management during pregnancy:

4. **Shearing:** **before lambing by 3-4 weeks**, in mild weather, this helps in:

1. Easier prediction of parturition by appearance of ewe
2. Help lamb to find teats and prevent wool ball formation in lamb rumen.
3. In bad weather force the ewe for more shelter seeking behavior.

5. **Vaccination:** against enterotoxaemia at **6-8 and 2-4 weeks prior** to lambing.

Signs of approaching lambing:

- 1. Lamb Stealing:** Before parturition 20% of ewes shows early maternal behavior resulted in lamb stealing “**Ewe steal a lamb of other ewe**”.
- 2. Penned sheep shows restlessness, circles, vocalizes, and paws at the ground 60-90 minutes before parturition.**
- 3. Ewe ceases grassing and rumination**

Signs of approaching lambing:

4. Shelter Seeking: A few days before parturition the ewe withdraws from the flock, and seeks some sort of shelter. In a pen she will seek a corner.

Shelter seeking by the ewe improves the environment into which the lamb is born so that its chances of **survival are greater**, but the ewe is responding primarily to her own **thermo-regulatory needs**.

Signs of approaching lambing:

5. She will **sniff and lick** at bedding contaminated with amniotic fluid.
6. Relaxation of **pelvic ligaments** and dropping in the **flank region**.
7. Increase in the **size of udder**.

Lambing or parturition:

1. The onset of parturition is signal by the **protrusion of the amniotic sac and part of the fetus**, ewe **lying down** and paddling with the hind feet and scraping the ground with foot.
2. The **older ewe spent shorter time** between onset of restlessness and labor, usually **30 to 60 minutes**. The peak lambing frequencies noted at **9 to 12 AM** and at **3 to 6 PM**.

Factors prolong the duration of the birth process:

- 1. When the lamb is relatively large for the birth canal.**
- 2. Ewes lambing for the first time and those are older than the usual.**
- 3. When the fetus takes up an abnormal posture during delivery.**
- 4. Exhaustion of the ewe.**

Expulsion of fetal membranes and placenta: occurs within 2 hr.

Licking or maternal care:

- -Maternal behavior in sheep has an important clinical aspect since most lamb **mortality** occurs within the **first week** of life in range-reared sheep. Mortality rates are from **5 to 15 % rising to as 50% in huge flock in bad weather**. Abnormal or weak maternal behavior accounts for the losses and increase mortality rate.
- Most **maternal rejection** or simply poor mothering without absolute rejection occurs most frequently in **young ewes and in those that had difficult at parturition**.

Licking or maternal care:

After lambing:

the ewe begins to **lick** her lamb and emit a **special parturition call**. If she stops licking, the lamb gives distress calls.

Licking is important for:

1. Establishment of **maternal bond** where, the ewe will be able to identify her lambs by **smell and taste**.
2. Licking of the lamb can be very important in cold or windy weather because it serves to **dry the neonate**.
3. Stimulates **lamb respiration**.
4. Stimulates of **blood and lymph circulation**.
5. Stimulation of **muscular tone and help him to stand**.
6. Stimulation of **urination and defecation**.

Licking or maternal care:

b. During suckling:

Once the lamb stands, usually within **30 minutes**, and begins to suckle the ewe continues to lick it, but mostly on the hindquarter to stimulate urination and defecation.

Acceptance of lamb:

- 1. The critical period during which ewe will **accept** the lamb is the first **5 min.** after parturition. Normally ewe will stay within **2 meters** of its lamb for most of the first day.
- 2. If an ewe's lamb is **removed immediately** after birth and before she has licked it, the ewe will **accept any** lamb presented to her **but once she has spent 20 to 30 minutes licking a lamb, her own or a substitute, she will not accept another.**
- 3. If **too long time** elapsed between parturition and lick of lamb, she may **reject him.**
- 4. Ewe temporarily accepts lamb rubbed with her **placenta.**

B. Goats:

1. Maternal behavior in goats is **similar** to that in sheep; but does seek more **isolated places for parturition**.
2. During the **1st 3 days** of the kid's life, it **does not follow its mother, but remains hidden**. After that it follows the doe.
3. Kids removed at birth and presented to the doe an **hour later may be rejected**. It is particularly important that the kid has no contact with other goats before accepted by its mother as it will reject a kid that smells of other goats.

B. Goats:

- 4. **5 min. of contact at birth is enough to form the bond**, approximately the same length as in sheep. **Olfaction** seems to be important, local anesthesia of the olfactory receptors of goats caused them to accept all kids, rather than only their own.
- 5. The small ruminants seem to be **unable to distinguish between species**. So that **lambs** can easily **cross -fostered to goats** and **vice versa**.

Management of ewe after lambing:

- 1. If expulsion of placenta is **delayed**, **veterinary** help should be called.
- 2. **Wash the udder and hindquarters** with lukewarm water containing antiseptic and dry with clean towel.
- 3. Do not start feeding grain to the ewe too soon after lambing. This may stimulate too much milk production and cause digestive problems. Let her have **3 or 4 pounds of hay or other roughage** for the first 24 hours.

Management of ewe after lambing:

- 4. The ewe will need some **water** after parturition.
- 5. **After 24 hours** have passes, the ewe can be offered some **grains**, the amount of which should be increased gradually until she is at full feed after the second day.
- 6. **Milking**: They must be milked at regular **12 h intervals**. Milking must be done away from the goat pen and any source of dust or odor. Cares before milking such as **clipping, washing udder, dryness of udder and disinfection after milking** must be considered.

Neonatal behavior:

- a. Suckling or nursing:
- 1. Although lambs are able to stand within one hour of birth, it may take 2 to 3 hours before they find the udder. She may either facilitate or inhibit teat seeking.
- 2. Once suckling has begun, it occurs with great frequency (twin lambs suckle 22 times during 16 hours of daylight, and single lambs 6 to 14 times) and duration as long as 3 min. / bout; but decreased with age.
- 3. During the first 2 weeks the ewe will allow one to suckle without the other. Later the ewe will refuse to let one nurse until the other is present.

Neonatal behavior:

- 4. If the ewe is **lying down**, the lambs will not only nudge her, but also will **jump over** her back and paw at her in an attempt to make her stand.
- 5. As **suckling decreases**, **grazing** by the lambs' **increases** and the lambs begin to form groups of their own and by **day 29 will spend 60% of their time with other lambs.**

b. Imprinting:

- The process of imprinting take place in a comparatively short period of time sometimes referred to as "**critical period**"
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- Critical period in sheep normally **take place a few hour following parturition**. The establishment of imprinting is though to be less a pure by **visual mechanisms**, than an **olfactory** one where the young animal is able to absorb.
- Studies have shown that, if a new born lamb is taken from the ewe **an hour** or so after birth & **returned after 8 hours**, the young lamb will still be **accepted** by the mother.

b. Imprinting:

- It is important that the dam involved is in the **critical period** & that by **licking & grooming** of the neonate its **specific odor** becomes **identifiable** to her, thereby **establishing the attachment**.
- However it has been found that if the **neonate lamb** is **removed immediately after birth** & **returned to the ewe** within a period of **2-4 h.**, it is likely to be **totally rejected**.

Management of lamb after birth:

- **1. Cut the umbilical cord:** at a distance of 5 cm. from the lamb's abdomen and disinfect it with iodine after tying it tightly.
- **2. Drying of lamb:** as soon as possible, dry the lamb from the fetal fluids and mucous by a piece of cloth or some straw especially the natural openings until it begin respiration normally.
- **3. Helping lamb to stand:** Aid the lambs to stand and to approach his mother and in weak lambs strip the teats in your hand train the lamb to lick it, then he will be able to locate his mother.

Management of lamb after birth:

- **4. Check the newborn for supra-mammary teat or pea-like growth at the tip of vulva which indicate hermaphrodite.**
- **5. Feeding of lamb:** her milk will be sufficient to meet the requirement of her offspring. At about **2 weeks**, the lambs eat **roughage** by placing some **hay** and after **4-5 weeks** they can feed **whole grains**.
- **6. The weaning of lambs occurs at 12-14 weeks of age.**

Feeding of orphan lamb:

- When a mother **dies** after parturition or **refuse** to secrete colostrum due to **inflammation** or **sensitivity** of the teats one of the following choice can be used:
 1. **Persuading ewe to accept lamb**
 2. **Grafting lamb on a different ewe (Foster mother).**
 3. **Artificial feeding or bottle feeding.**